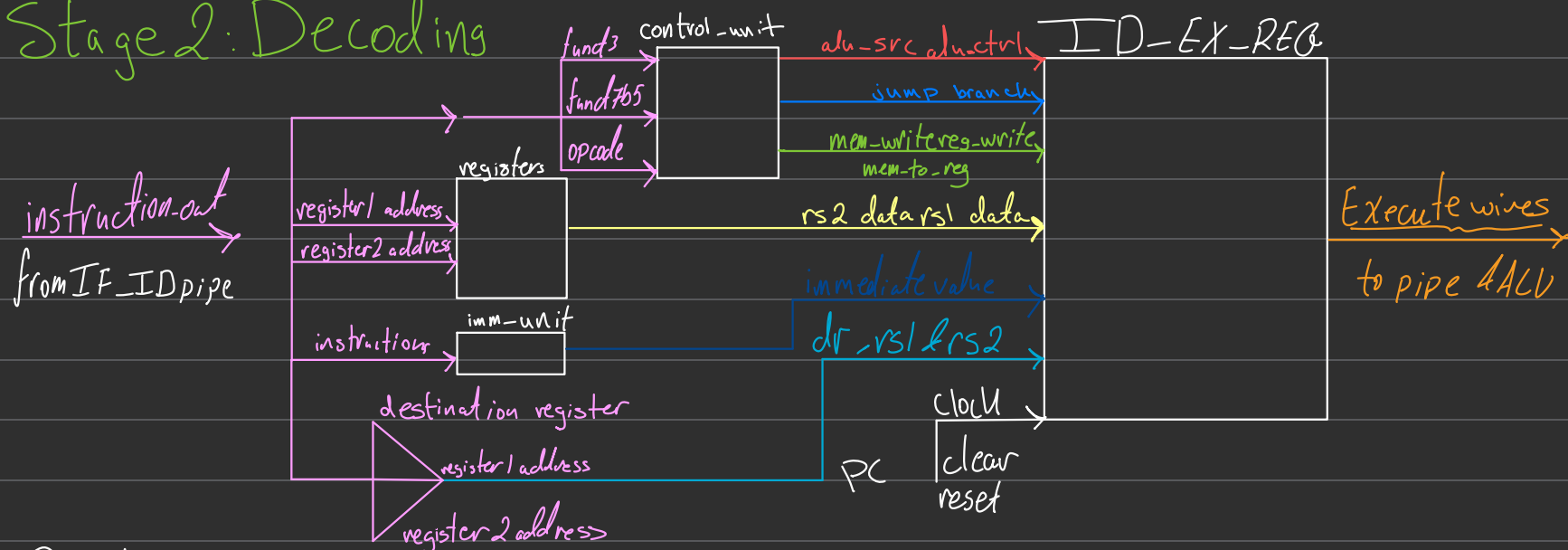


CPU Architecture 1.0

Stage 2: Decoding



Control unit:

Input:

opcode: identifies the broad instruction category: ALU operation, jump & so on.

funct3: 3-bit function that is a bit more specific about what it needs.

funct7b5: extra bit for funct3.

Output:

alu-ctrl: 4-bit instruction to do Add, subtract or MUX, etc.

alu-src: 1-bit switch for the ALU's MUX $\begin{cases} 0, \text{if R-type} \\ 1, \text{if I-type} \end{cases}$

mem-to-reg: 1-bit switch that chooses what data is written onto the register destination (dr) $\begin{cases} 0, \text{ALU result} \\ 1, \text{ram memory} \end{cases}$

mem-write: 1-bit switch that chooses if & what data is written onto the memory $\begin{cases} 0, \text{no change} \\ 1, \text{write the values of rd2 onto the specified memory address} \end{cases}$

reg-write: 1-bit switch that chooses if the dr will change $\begin{cases} 0, \text{no change} \\ 1, \text{save the data on data @} \end{cases}$ $\begin{cases} 0, \text{no change} \\ 1, \text{write the values of rd2 onto the specified memory address} \end{cases}$